Genesis Protect Acute v1

Claim Verification Map

On-Chain/Off-Chain Boundary · PHI Separation · LP Capital Protection

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Status	Pre-Mainnet Review
Audience	Investors · LPs · Sponsors · External Auditors
Classification	Internal — Pre-Mainnet Analysis
Network	Solana (Devnet at publication)

Three Protocol-Level Guarantees — No Operator Trust Required

1. Reserve capital cannot leave the vault without a valid, attested, settled ClaimCase
2. Evidence integrity is tamper-evident — hash is permanently immutable once the oracle attests
3. Raw patient data (PHI) never appears on Solana in any form — only 32-byte SHA-256 hashes

Every claim state · Every verification event · Every hash anchor · Fully auditable from public chain data

Executive Summary

Genesis Protect Acute v1 processes health insurance claims through a cryptographically verifiable truth chain anchored on Solana. This document is the authoritative map of that chain — for investors, LPs, sponsors, and auditors who need to understand what the protocol guarantees and how to verify those guarantees from public on-chain data, **without accessing any patient health records**.

Three properties are enforced at the protocol level — no operator trust required:

#	Property	How it is enforced
1	Reserve capital cannot leave the vault on an unauthorized claim	Only outflow path: PDA-signed SPL transfer gated on ClaimCase.intake_status = SETTLED (4) with a valid oracle attestation chain
2	Claim evidence is tamper-evident — retrospective forgery is impossible	evidence_ref_hash immutable once attestation_count ≥ 1; hash equality enforced on-chain at attest_claim_case
3	Raw patient data (PHI) never appears on Solana in any form	Only 32-byte SHA-256 fingerprints written on-chain; raw documents stay encrypted on the off-chain oracle portal

What LPs and sponsors can verify independently — from public Solana explorer data alone:

- Every claim opened, every evidence hash committed, every oracle attestation, every adjudication decision, and every payout — each with timestamp and signing authority
- That vault balances decrease only on settled, attested claims
- That oracle attestations reference exactly the evidence packet that was reviewed (hash equality enforced on-chain — transaction reverts on mismatch)
- That protocol fees were correctly carved out at settlement

> **Phase 0 posture:** Operator-backed oracle review with all material state transitions anchored on-chain. The roadmap toward fully decentralized adjudication is documented in §10.

§1 — Legend & Pre-Claim Foundation

Symbol	Meaning
■	Solana instruction (on-chain transaction)
■	On-chain proof anchor (hash or state written to Solana)
■	Off-chain only (raw PHI, operator workflow, AI logs)
■	TEE only (MagicBlock private review — hash exits, plaintext never does)

Before any claim opens, the protocol anchors the policy terms, pricing, and evidence requirements on-chain as **immutable hash commitments**. All material is locked (material_locked = true) before the series goes live — every claim is evaluated against exactly these anchored parameters.

PolicySeries account — set by protocol governance:

Field	What it commits to
`terms_hash`	Full coverage terms and exclusion schedule
`pricing_hash`	Premium structure and benefit tiers
`payout_hash`	Benefit amounts per tier (T1/T2/T3) and reimbursement cap
`reserve_model_hash`	Reserve methodology and VaR parameters
`evidence_requirements_hash`	Required document types per claim tier

HealthPlan account — set by plan admin:

Field	What it commits to
`schema_binding_hash`	Binding to the verified OutcomeSchema for this plan

*The **OutcomeSchema** (genesis-protect-acute-claim v1) is separately registered on-chain and must be marked verified = true by governance before any oracle can attest. Its schema_key_hash and schema_hash are snapshotted into every ClaimAttestation.*

§2 — Claim State Machine

The ClaimCase account's intake_status field drives the complete claim lifecycle. No state can be skipped or forged without a valid signed Solana transaction from the authorized role.

> **Note on the DENIED → OPEN arrow:** after denial, a new ClaimCase PDA is opened for an appeal — this is not an in-claim state rollback. The original ClaimCase remains permanently at DENIED (3) on-chain.

State	intake_ status	On-chain trigger	Who signs	Money moves?
OPEN	`0`	open_claim_case	Member or claims_operator or plan_admin	No
UNDER_REVIEW	`1`	Operator queue pickup (off-chain)	—	No
APPROVED	`2`	adjudicate_claim_case (approved_amount > 0)	claims_operator	No — reserve booked
DENIED	`3`	adjudicate_claim_case (approved_amount == 0)	claims_operator	No — obligation void
SETTLED	`4`	settle_claim_case / settle_claim_case_selected_asset	claims_operator	**YES — USDC exits vault**
CLOSED	`5`	Reserved — Phase 1 dispute-case state	—	—

§3 — Instruction-to-Verification Map

Every step that produces an on-chain proof is listed with the exact account written, the hash anchored, and the event emitted. Off-chain steps (■) are shown for completeness — they leave no on-chain footprint at that point.

Step 1 — open_claim_case ■

Authorized signers	Member wallet OR health_plan.claims_operator OR health_plan.plan_admin
PDA created	ClaimCase — seeds: ["claim_case", health_plan.key(), claim_id]
State transition	→ CLAIM_INTAKE_OPEN (0)
■ On-chain writes	claim_case.claimant · .policy_series · .funding_line · .evidence_ref_hash (initial) · .opened_at
Event emitted	ClaimCaseStateChangedEvent { claim_case, intake_status: 0, approved_amount: 0 }
■ Off-chain at this point	Raw incident report, member identity docs — not yet submitted
PHI on-chain?	NO
■ Security (PT-04)	claimant constrained to member_position.wallet — operators cannot spoof claimant

Step 2 — authorize_claim_recipient ■ (optional)

Authorized signers	Member wallet ONLY — operators cannot set this
State transition	None
■ On-chain writes	claim_case.delegate_recipient (payout routing address)
Event emitted	None
PHI on-chain?	NO

Step 3 — Evidence upload & human review ■

Where	OmegaX Health oracle portal (off-chain, encrypted)
■ Member uploads	Discharge summary · itemized invoice · location proof · doctor note
■ AI pre-screening	Completeness check, anomaly flags → operator queue
■ Human review	Claims operator reviews raw PHI against policy_series.evidence_requirements_hash
PHI on-chain?	NO — raw documents never touch Solana

Step 4 — attach_claim_evidence_ref ■

Authorized signers	health_plan.claims_operator
Precondition	claim_case.attestation_count == 0 (mutable only before first oracle attestation)
State transition	None
■ On-chain writes — hash 1	claim_case.evidence_ref_hash [SHA-256 of full evidence packet]
■ On-chain writes — hash 2	claim_case.decision_support_hash [SHA-256 of operator review bundle]
Event emitted	None
PHI on-chain?	NO — only 32-byte hashes
Evidence lock	Once attestation_count ≥ 1: evidence_ref_hash permanently immutable. Revised evidence → new ClaimCase PDA.

Step 5 — attest_claim_case ■

Authorized signers	oracle_profile.oracle
Preconditions	outcome_schema.verified == true · oracle has ATTEST_CLAIM permission · args.attestation_ref_hash == claim_case.evidence_ref_hash (enforced on-chain — reverts on mismatch) · finality hold clear
PDA created	ClaimAttestation — seeds: ["claim_attestation", claim_case.key(), oracle.key()]
State transition	claim_case.attestation_count++
■ On-chain writes	Full ClaimAttestation — see §3.5 field breakdown below
Event emitted	ClaimCaseAttestedEvent { claim_attestation, claim_case, oracle_profile, oracle, decision, attestation_hash }
PHI on-chain?	NO — only hash fields
■ Security (PT-07)	Oracle registration requires signer == args.oracle — no third-party registration possible

Step 6 — adjudicate_claim_case ■

Authorized signers	health_plan.claims_operator
State transition (approve)	→ APPROVED (2) if approved_amount > 0
State transition (deny)	→ DENIED (3) if approved_amount == 0
■ On-chain writes	claim_case.intake_status · .adjudicator [Pubkey] · .approved_amount · .denied_amount · .decision_support_hash
Event emitted	ClaimCaseStateChangedEvent { claim_case, intake_status, approved_amount }
Denial path	No Obligation PDA created. Denial reason anchored via decision_support_hash — full rationale off-chain.
PHI on-chain?	NO — only amounts and pubkeys

Step 7 — reserve_obligation ■

Authorized signers	health_plan.claims_operator
State transition	Obligation.status → RESERVED (1)
■ On-chain writes	obligation.reserved_amount · funding_line.reserved_amount ↑ · claim_case.linked_obligation
Event emitted	ObligationStatusChangedEvent { obligation, funding_line, status: 1, amount }
Visibility	Encumbered reserve is visible on the public protocol console in real time
PHI on-chain?	NO

Step 8 — settle_claim_case / settle_claim_case_selected_asset ■

Authorized signers	health_plan.claims_operator
Preconditions	reserve_asset_rail.payout_enabled == true · price freshness within rail bounds · recipient token account owner matches resolved recipient
State transition	→ SETTLED (4) when paid_amount ≥ approved_amount
■ On-chain writes	claim_case.paid_amount ↑ · .intake_status → SETTLED · .closed_at · domain_asset_vault.total_assets ↓
SPL transfer	transfer_from_domain_vault (PDA-signed CPI) — THE ONLY AUTHORIZED OUTFLOW PATH
Events emitted	ClaimCaseStateChangedEvent + FeeAccruedEvent (protocol fee) + optional FeeAccruedEvent (oracle revshare if LP-backed)
Multi-asset variant	settle_claim_case_selected_asset → ClaimCaseSelectedAssetPayoutEvent { claim_asset_mint, payout_asset_mint, claim_credit_amount, payout_amount, settlement_reason_hash }
PHI on-chain?	NO
■ Security (PT-01/02)	transfer_from_domain_vault requires vault PDA signature via seeds — no unsigned outflow possible

ClaimAttestation — Complete On-Chain Fields (from Step 5):

Field	Size	What it represents
`oracle`	Pubkey	Oracle authority public key
`oracle_profile`	Pubkey	OracleProfile PDA reference
`claim_case`	Pubkey	Parent ClaimCase reference
`health_plan`	Pubkey	HealthPlan reference
`policy_series`	Pubkey	PolicySeries reference
`decision`	u8	0=approve / 1=deny / 2=request_review / 3=abstain
`attestation_hash`	32 bytes	Oracle's own hash commitment (off-chain signed artifact)
`attestation_ref_hash`	32 bytes	MUST equal claim_case.evidence_ref_hash — enforced on-chain
`evidence_ref_hash`	32 bytes	Snapshot of evidence hash at attestation time
`decision_support_hash`	32 bytes	Snapshot of operator review bundle hash
`schema_key_hash`	32 bytes	Hash of the OutcomeSchema key ("genesis-protect-acute-claim")
`schema_hash`	32 bytes	Hash of the schema content at attestation time
`schema_version`	u16	Schema version at attestation time
`created_at_ts`	i64	Unix timestamp of attestation

§4 — PHI vs. Proof Anchoring: Complete Separation

This table is the canonical reference for what lives where. Nothing in the left column ever appears on Solana in any form. The right column is what external parties can read and verify.

Category	Data (■ off-chain only)	■ On-chain proof anchor
Member identity	Passport, ID document, KYC data	`member_position.subject_commitment [identity commitment hash]`
Medical evidence	Discharge summary, doctor notes, clinical records	`claim_case.evidence_ref_hash [SHA-256 of evidence packet]`
Billing documents	Itemized invoice, proof of payment, receipts	`claim_case.evidence_ref_hash (included in evidence packet)`
Operator review	Review checklist, internal notes, AI flag log	`claim_case.decision_support_hash [SHA-256 of review bundle]`
Oracle decision	Full oracle assessment document	`claim_attestation.attestation_hash [oracle's 32-byte commitment]`
Policy terms	Full coverage terms text (PDF / doc)	`policy_series.terms_hash [SHA-256 of terms document]`
Evidence requirements	Required document checklist per tier	`policy_series.evidence_requirements_hash`
Fraud notes	Investigation notes, referral details	*Not anchored on-chain in Phase 0*
AI screening logs	Pre-screening output, anomaly scores	*Not anchored on-chain in Phase 0*
TEE evidence packet	Plaintext PHI inside MagicBlock TEE (never exits)	`private_review_session.review_artifact_hash (exits TEE as hash only)`

> **Summary:** Every byte of raw PHI stays off-chain. What Solana records are the SHA-256 fingerprints of that data — sufficient for a third party to verify that the reviewed evidence matches what was attested, **without accessing the underlying medical content**.

§5 — Complete On-Chain Hash Chain

The chain below shows how each hash links to the next, forming a tamper-evident path from raw PHI to final USDC settlement on Solana.

RAW PHI (off-chain only)

`Medical records + invoices + location proof + doctor note Stored encrypted on OmegaX Health oracle portal`

▼ *SHA-256 by operator after human review*

■ ClaimCase.evidence_ref_hash [32 bytes — Solana]

`Set by: attach_claim_evidence_ref (claims_operator) Immutable after: first attest_claim_case (attestation_count ≥ 1)`

▼ *Oracle verifies: attestation_ref_hash == evidence_ref_hash*

Enforced on-chain — transaction reverts if hashes do not match

■ ClaimAttestation PDA [Solana — permanent]

`attestation_hash · attestation_ref_hash · evidence_ref_hash decision_support_hash · schema_key_hash · schema_hash decision [u8] · oracle [Pubkey]`

▼ *adjudicate_claim_case (claims_operator)*

■ ClaimCase — adjudication record [Solana — permanent]

intake_status: APPROVED (2) or DENIED (3) adjudicator [Pubkey] · approved_amount [u64] · decision_support_hash

▼ *reserve_obligation → settle_claim_case*

■ Settlement — SPL token transfer [Solana — permanent]

intake_status: SETTLED (4) · paid_amount [u64] · closed_at [i64] recipient: member_position.wallet or delegate_recipient domain_asset_vault.total_assets ↓ · tx signature immutable

Verification property: *A third party who holds the original evidence packet can SHA-256 hash it and compare against claim_case.evidence_ref_hash. If they match, the evidence anchored on Solana is identical to what was reviewed — and the oracle's attestation applies to exactly that packet.*

§6 — MagicBlock TEE Private Review Path

For claims requiring TEE-private handling, an adjunct program omegax_private_claim_review runs on MagicBlock Ephemeral Rollups. This path integrates at steps 4–5 of the standard lifecycle. **The main protocol vaults, funding lines, and obligations are never delegated to MagicBlock.**

Standard path	MagicBlock TEE private path
■ Evidence upload (off-chain)	■ Evidence upload (off-chain)
■ attach_claim_evidence_ref	■ TEE receives private packet ■ open_review_session
■ attest_claim_case	■ delegate_review_session (→ MagicBlock ER) ■ Private review inside TEE — plaintext never leaves enclave ■ record_private_review (review_artifact_hash only) ■ record_private_payment_ref (payment_reference_hash only) ■ commit_and_close_review_session (→ Solana base) ■ attest_claim_case (main program — consumes artifact hash)

PHI boundary in the MagicBlock path:

What	Where
Private evidence packet (plaintext)	■ TEE enclave only — never on Solana, never on MagicBlock ER
review_artifact_hash	■ MagicBlock ER → committed to Solana base
private_payment_reference_hash	■ MagicBlock ER → committed to Solana base
Oracle attestation hash	■ Standard ClaimAttestation PDA on Solana base

§7 — Unhappy Paths: Verification Map

Scenario	On-chain state	What's verifiable
Incomplete evidence	intake_status = OPEN (0), no evidence_ref_hash set	Chain shows claim opened but no evidence attached
Evidence resubmission (pre-attestation)	New attach_claim_evidence_ref updates evidence_ref_hash	Each update is a distinct Solana tx; full history recoverable from tx logs
Evidence revision (post-attestation)	New ClaimCase PDA opened	Both cases publicly visible; original attestation chain immutable
Partial approval (Travel 30 top-up)	approved_amount = fixed tier cap; second adjudication round for top-up	Two separate ClaimCaseStateChangedEvent entries on-chain
Denial	intake_status = DENIED (3); decision_support_hash updated	Hash-committed reason; no obligation created; no funds reserved
Settlement deferred	obligation.status = CLAIMABLE_PAYABLE (2)	Encumbered reserve visible; vault balance unchanged
Impairment	obligation.status = IMPAIRED (5) after mark_impairment	ImpairmentRecordedEvent on-chain; LP junior class absorbs loss
Appeal (Phase 0)	New ClaimCase PDA with appeal evidence	New claim on-chain; original denial chain intact and immutable

§8 — Obligation State Lifecycle

The Obligation account runs in parallel with ClaimCase. Every status transition emits ObligationStatusChangedEvent \{ obligation, funding_line, status, amount \}.

Status	Value	Trigger	Effect
PROPOSED	0	adjudicate_claim_case (approved)	Obligation PDA created; no reserve yet
RESERVED	1	reserve_obligation	funding_line.reserved_amount ↑ — LP capacity encumbered
CLAIMABLE_PAYABLE	2	Internal transition	Ready for settlement payout
SETTLED	3	settle_claim_case	Funds disbursed; obligation closed
CANCELED	4	From RESERVED or PROPOSED — denial / void	No reserve taken; obligation void
IMPAIRED	5	mark_impairment — from RESERVED or CLAIMABLE_PAYABLE	LP junior class absorbs loss; ImpairmentRecordedEvent emitted
RECOVERED	6	Post-impairment recovery — from IMPAIRED	Partial or full recovery credited back to LP

§9 — Independent Auditor Verification Checklist

For any Genesis Protect Acute claim, an independent auditor can verify the following from **public on-chain data alone**. No access to raw PHI is required.

Question	Where to look on-chain
Who opened the claim and when?	`claim_case.claimant · .opened_at · opening tx signature`
Was the claimant an active policy member?	`member_position.active == true · .eligibility_status == ELIGIBLE at claim time`
What evidence hash was committed?	`claim_case.evidence_ref_hash`
Was evidence locked before attestation?	`claim_case.attestation_count ≥ 1 → hash is immutable`
Who attested and with what decision?	`ClaimAttestation.oracle · .decision · .attestation_hash`
Was the schema governance-verified?	`outcome_schema.verified == true at attestation time`
Does the attestation match the evidence?	`claim_attestation.attestation_ref_hash == claim_case.evidence_ref_hash (enforced on-chain)`
Who adjudicated and what was the outcome?	`claim_case.adjudicator · .intake_status · .approved_amount`
What funding line was reserved?	`claim_case.funding_line · obligation.reserved_amount`
Was payout made and to whom?	`claim_case.paid_amount · .closed_at · settlement tx · recipient = delegate_recipient or member_position.wallet`
Were fees correctly carved out?	`FeeAccruedEvent on settlement tx · protocol_fee_vault.accrued_fees`
Was policy material unchanged at claim time?	`policy_series.material_locked == true · terms_hash unchanged since issuance`

§10 — Phased Decentralization Roadmap

The protocol launches in a deliberately phased approach: Phase 0 achieves on-chain anchoring of all material claim state while retaining operator-supervised workflows for review queuing and oracle key management. The table below is an honest accounting of what each phase adds.

Item	Phase 0 posture	Phase 1 target
UNDER_REVIEW state	Off-chain queue pickup — not a distinct on-chain intake_status transition	Explicit on-chain set_claim_under_review instruction
Appeal linkage	New ClaimCase PDA for appeals; correlated off-chain	On-chain dispute-case linkage between original denial and appeal PDA
Fraud referral record	Internal notes; hash not yet committed on-chain	fraud_referral_hash field or dedicated instruction in protocol-oracle-service
Oracle key custody	Claims operator hot wallet signs oracle transactions	HSM / KMS boundary enforced before mainnet (see claims-processing-spec §13)
Provider direct settlement	Payout goes to member wallet or delegated recipient	Provider KYB + delegate_recipient = verified provider wallet

Phase 0 communication standard: Any external claim communication — to members, partners, or press — must describe the current posture as **operator-backed oracle review**, not fully decentralized adjudication. The on-chain record is authoritative; the governance of who writes to it is progressively decentralizing.

§11 — LP Capital Protection Summary

For LPs deploying capital into OmegaX Protocol liquidity pools, the claim architecture provides the following hard guarantees at the smart contract level.

LP Concern	Answer	Detail
Can an operator drain the vault without a real claim?	**NO**	Funds leave only via transfer_from_domain_vault PDA-signed CPI, gated on ClaimCase.intake_status = SETTLED (4) with a valid oracle attestation chain
Can a fraudulent claim override a valid attestation?	**NO**	adjudicate_claim_case is gated on the oracle attestation chain; adjudicator pubkey permanently recorded on-chain
Can evidence be altered after oracle review?	**NO**	evidence_ref_hash is immutable once attestation_count ≥ 1; any revision requires a new ClaimCase PDA
Can LP losses be hidden?	**NO**	ImpairmentRecordedEvent emitted on-chain; ObligationStatusChangedEvent at every obligation state transition
Can payout routing be changed by the operator?	**PARTIALLY**	Operator cannot override a member-set delegate_recipient; authorize_claim_recipient is member-only
Is the oracle schema governance-controlled?	**YES**	outcome_schema.verified == true enforced on-chain at attestation; schema changes require governance authority signature

§12 — References

Document	Path
Step-by-step claim truth chain narrative	`docs/architecture/genesis-protect-claim-trace.md`
Claims processing specification (KR 1.2)	`docs/architecture/genesis-protect-acute-claims-processing-spec.md`
Full operational protect flow (KR 1.1)	`docs/architecture/genesis-protect-acute-full-protect-flow.md`
MagicBlock private claim room	`docs/architecture/magicblock-private-claim-room.md`
Evidence schema (JSON)	`frontend/public/schemas/genesis-protect-acute-claim-v1.json`

Anchor source — claim instructions	`programs/omegax_protocol/src/claims.rs`
Anchor source — account states	`programs/omegax_protocol/src/state.rs`
Anchor source — protocol events	`programs/omegax_protocol/src/events.rs`
Anchor source — constants and seeds	`programs/omegax_protocol/src/constants.rs`